Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system under nonnegative adversarial arrivals, convex state–action costs, and state-dependent action constraints. At each time, the controller chooses a feasible action before the current arrival and cost are revealed. We introduce *Simplex Disturbance-Action Control* (SDAC), a simplex-constrained finite-memory policy class whose fixed policies satisfy the per-step action constraint by construction. For online adaptation, an entropic online mirror descent (OMD) update maintains the SDAC parameters on the subprobability simplex, while scalar action clipping ensures feasibility of the actual time-varying trajectory. We show that this scalar clipping is exactly the action induced by an entropic Bregman projection onto the current availability constraint, although the projected parameter need not be computed. We prove $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy, with a leading constant that is $\Theta(\alpha)$ as $\alpha \downarrow 0$ and $\Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$, where H is the memory horizon. We also prove a minimax lower bound that matches the small-α power. Once T is at least the mixing scale $\Theta((1-\alpha)^{-1})$, the lower bound is $\Omega((1-\alpha)^{-1}\sqrt{T})$ as $\alpha \uparrow 1$, leaving a square-root gap in the stability-margin dependence of the upper and lower bounds. We further show that SDAC geometrically approximates every feasible proportional state-feedback policy and a broader class of decreasing allocation-sequence policies motivated by energy-harvesting battery management. Choosing $H = \Theta(\log T)$ yields $O\left(\sqrt{T\log\log T}\right)$ regret against the best policy in either class. The setting also applies to related storage systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY dynamic resource systems can be modeled by a storage state that evolves over time and constrains the control action at every time. The controller must choose an action subject to a state-dependent action constraint before observing the corresponding arrival and cost.

This creates a constrained discrete-time control problem with state–action coupling: aggressive early actions can harm later performance, while overly conservative actions can increase the stage cost. Our objective is to remain feasible at every time while competing with the best fixed policy in a tractable class that approximates feasible proportional state-feedback policies.

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B. Model and Information Structure

We study a discrete-time storage system with state $x_t \geq 0$, where x_t models the storage level at the beginning of time t. At each time t, the controller chooses a control action u_t under the state-dependent action constraint

$$0 \leq u_t \leq \alpha x_t,$$

where $\alpha \in (0,1)$ is a fixed known retention coefficient (so a fraction $1-\alpha$ of the state is lost each period). After the action is chosen, a nonnegative adversarial arrival w_t (a resource inflow) is revealed, and the state evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t.$$

Although this input is commonly called a disturbance in the online-control literature, we call $w_t \in [0,1]$ an *arrival* to emphasize its nonnegativity; we retain the standard names Disturbance-Action Control (DAC) and Simplex Disturbance-Action Control (SDAC). Nonnegativity is essential to the storage interpretation and to the feasibility properties of SDAC.

Unlike stochastic models, our formulation is adversarial and non-anticipatory. At time t, the environment may select w_t and c_t using the history through time $t-1$, but not the current action; the controller then chooses u_t, after which w_t and the entire function c_t are revealed. The goal is low cumulative cost relative to the best fixed policy in a class defined below, with both the online and benchmark trajectories remaining feasible.

C. Application Domains

This abstraction captures a broad set of domains:

- **Energy harvesting and battery management:** x_t is battery state of charge (retention coefficient α after self-discharge), w_t is harvested energy, and u_t is discharge or consumption bounded by available energy αx_t.
- **Queueing networks and buffer management:** x_t is queue/buffer occupancy, w_t is packet arrivals, and u_t is service bounded by αx_t.
- **Inventory control:** x_t is on-hand inventory (retention coefficient α after spoilage/deterioration), w_t is supply or replenishment, and u_t is sales bounded by αx_t.
- **Water-reservoir management:** x_t is stored water (retention coefficient α after evaporation/seepage), w_t is inflow, and u_t is controlled release bounded by αx_t.

We take energy harvesting as the running interpretation; the other domains share the same storage dynamics and state-dependent action constraint.

D. Contributions

Our contributions are as follows:

- **Policy class.** We introduce SDAC, a DAC-inspired policy class for storage dynamics with the state-dependent action constraint $0 \leq u_t \leq \alpha x_t$. We prove that every fixed SDAC policy is feasible, its induced loss is convex and Lipschitz in the policy parameter, and the class geometrically approximates feasible proportional state-feedback policies and a broader class of decreasing allocation-sequence policies.
- **Online control and regret.** We develop an entropic OMD controller over SDAC parameters. A Bregman projection maintains the simplex constraint on the parameters, and scalar action clipping maintains feasibility of the actual trajectory. The clipping has an equivalent Bregman-projection interpretation, but its direct implementation requires only a scalar minimum. A sharp coordinatewise gradient estimate and a contractive trajectory comparison give $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy, with upper-bound dependence $\Theta(\alpha)$ as $\alpha \downarrow 0$ and $\Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$, and, with $H = \Theta(\log T)$, achieves $O\big(\sqrt{T\log\log T}\big)$ regret against the best feasible allocation-sequence policy, and hence also against the best proportional state-feedback policy.
- **Minimax lower bound.** We construct an oblivious sequence of arrivals and convex costs for which every causal feasible controller has expected SDAC regret $\Omega(G_\alpha\sqrt{T})$ after the mixing scale, where $G_\alpha = \alpha(L_x/(1-\alpha) + L_u)$. This matches the small-α power of the upper bound. As $\alpha \uparrow 1$, it gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ once $T = \Omega((1-\alpha)^{-1})$, leaving a factor $(1-\alpha)^{-1/2}$ (apart from the memory logarithm) to be closed.

E. Related Work

a) Online nonstochastic control: A recent line of work studies online control of linear dynamical systems under adversarial disturbances and costs. Its performance criterion is cumulative regret relative to a benchmark policy class chosen in hindsight, under suitable stability conditions [1]–[3]; see [4] for an overview.

A central idea in this literature is *Disturbance-Action Control* (DAC),[1] a policy parameterization that combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard assumptions, this representation yields tractable surrogate losses in the policy parameters and supports efficient no-regret updates. Extensions include partial observability [5] and bandit feedback [6].

Constrained online-control methods have also been developed for affine and general static state–input constraints [7], [8]. Our focus is the storage-specific constraint $0 \leq u_t \leq \alpha x_t$, whose feasible action set depends on the current state. Exploiting nonnegative arrivals, SDAC uses a simplex parameterization to make every fixed policy feasible; scalar action clipping preserves feasibility when the parameter changes online.

[1]"Disturbance-Action Control" is the standard name in the online-control literature. In this paper the underlying signal is the nonnegative arrival w_t; we retain the DAC/SDAC names for continuity with that literature.

Some lower-bound results in online nonstochastic control establish optimality in the horizon T without resolving all stability-parameter dependence; for example, Simchowitz et al. establish the optimal $\sqrt{T}$ horizon dependence for known systems in their fully adversarial setting [3]. Kumar et al. develop matching memory-dependent upper and lower bounds for online convex optimization with unbounded memory [9]. Their discounted-memory specialization has lower-bound order $\sqrt{T}/(1-\rho)$. This scaling motivates comparison with our large-α result, but their theorem does not directly imply ours: their loss may inspect separate coordinates of the decision history, whereas our stage cost observes only the scalar state–action pair (x_t, u_t) under the storage feasibility constraint.

b) Energy harvesting: Online energy-harvesting control commonly uses simple battery-responsive rules. Shaviv and Özgür [10] introduced a fixed-fraction power policy and established distribution-uniform additive and multiplicative performance guarantees for i.i.d. energy arrivals. Arafa et al. [11] extended this viewpoint to general concave utilities and, for Bernoulli recharges, characterized optimal post-recharge expenditure through a generally nongeometric allocation sequence. Allocation-sequence structure also appears in look-ahead power control: Zibaeenejad and Chen [12] characterize an optimal positive decreasing sequence between recharge epochs. These models reveal the current harvested energy before transmission and use stochastic arrivals and a finite battery that may saturate.

Motivated by this literature, we introduce a comparator that assigns every arrival cohort the same decreasing release sequence, shifted by one slot to respect our information pattern. The resulting policy permits nongeometric expenditure profiles while remaining feasible under leakage. Unlike the renewal models above, different arrival cohorts may overlap, arrivals are adversarial, and performance is measured by regret for general convex state–action costs. Section IV-D shows that finite-memory SDAC policies geometrically approximate this entire class.

Lin et al. [13] study online age-of-information optimization in an energy-harvesting system with related battery dynamics. They compare with an unrestricted offline optimum through competitive ratio; their objective depends on the energy allocation but not directly on the battery state, and the current input is revealed before allocation. In our model, the controller acts before observing the current arrival and cost, the cost may depend jointly on state and action, and performance is measured by regret against a fixed policy class.

Yu and Neely [14] study expected utility maximization under i.i.d. system states. Asgari and Neely [15] express the guarantee as an $O(\sqrt{T})$ regret–battery-capacity tradeoff and improve its dependence on the decision dimension, allowing arbitrary loss sequences while retaining i.i.d. energy arrivals. Both works use action-only objectives and fixed-decision comparators. Here, arrivals are adversarial, costs depend on both state and action, and the comparator is a fixed state-responsive policy.

c) Queueing and network optimization: Queueing control provides a mathematically related perspective based on stochastic arrivals and real-time service decisions. Foundational work emphasizes throughput and queue stability [16]–[18], while stochastic network optimization also studies trade-offs between time-average objectives and backlog [19]. Our setting differs in its adversarial arrivals, convex state–action costs, and regret benchmark against a policy chosen in hindsight.

In many queueing applications, the natural retention coefficient is $\alpha = 1$ (no abandonment or leakage), so those systems are not direct instances of the $\alpha \in (0, 1)$ model analyzed here. Section VII discusses the additional issues that arise when $\alpha \geq 1$.

Among other results, Khojastepour and Sabharwal [20] study a time-invariant robust scheduler representable as a linear filter of past arrivals. It is designed to minimize average transmit power subject to a maximum packet-delay guarantee, rather than updated online to minimize revealed general costs.

d) Inventory management: Yang et al. [21] study online procurement under inventory constraints. The controller observes the current demand and price, then procures supply while ensuring that demand is met; performance is measured by competitive ratio for aggregate procurement cost. Our controller instead chooses an outflow before observing the current inflow and cost, and permits convex state–action costs.

Hihat and Fermanian [22] consider an inventory-control formulation with potentially nonlinear and nondifferentiable state-transition functions and study base-stock policies. While that model is more general in scope, its decision structure differs from ours: as in [21], the controller chooses supply and then observes demand through the transition, whereas our controller chooses an outflow and then observes incoming supply. Our scalar linear recursion and simplex policy class yield explicit feasibility and regret guarantees; the two frameworks target different dynamics, decisions, and policy classes.

e) Water-reservoir management: Classical dam theory studies storage dynamics with random inflows and controlled releases (see, for example, [23], [24]). Relative to our setting, this literature assumes stochastic inputs and emphasizes stationary or long-run performance quantities. It provides historical context for storage control under state constraints.

F. Paper Organization

Section III formalizes the adversarial storage-control model, assumptions, and regret benchmark. Section IV introduces the SDAC policy class, its structural properties, and its approximation of proportional and allocation-sequence policies. Section V presents the online controller, Section VI develops regret upper and lower bounds, Section VII discusses possible extensions, and Section VIII concludes. Deferred proofs appear in the Appendix.

II. Notation

We write $[T] \triangleq \{1, \dots, T\}$ and $[z]_+ \triangleq \max\{z, 0\}$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is v_i; in particular, the coordinates of $\mathbf{m}_t$ and $\mathbf{g}_t$ are $m_{t,i}$ and $g_{t,i}$. We write $\mathbf{1}$ for the all-ones vector (dimension is clear from context). The Euclidean inner product is denoted by $\langle \cdot, \cdot \rangle$. We write $\log$ for the natural logarithm. We set $w_\tau \equiv 0$ for $\tau \leq 0$. A fixed SDAC policy with parameter $\mathbf{m}$ is written $\pi_{\mathbf{m}}$, with induced trajectories $x_t(\mathbf{m})$ and $u_t(\mathbf{m})$. A feasible proportional policy with gain k induces $x_t(k)$ and $u_t(k)$. A decreasing allocation sequence $\mathbf{q}$ induces $x_t(\mathbf{q})$ and $u_t(\mathbf{q})$. After the online algorithm is introduced, unadorned x_t and u_t denote the realized clipped trajectory.

III. Model and Problem Setup

Let T be a positive integer. The system state $x_t \in \mathbb{R}_{\geq 0}$ denotes the resource level at the beginning of time t. The initial state is given as

$$x_1 = 0. \tag{1}$$

Fix a retention coefficient $\alpha \in (0, 1)$. At each time $t \in [T]$, the controller chooses a control action $u_t \in \mathbb{R}_{\geq 0}$ subject to the state-dependent action constraint

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

For every feasible trajectory,

$$0 \leq x_{t+1} \leq \alpha x_t + 1.$$

Since $x_1 = 0$, induction gives

$$0 \leq x_t \leq \frac{1 - \alpha^{t-1}}{1 - \alpha} \leq \frac{1}{1 - \alpha}, \qquad t \in [T + 1].$$

Thus, the storage level is uniformly bounded over time. In particular, a storage device with capacity $1/(1 - \alpha)$ suffices to realize every feasible trajectory, including the trajectory generated by our online controller, without overflow. Thus, every feasible state–action pair belongs to the compact set

$$\mathcal{D}_\alpha \triangleq \left\{ (x, u) : 0 \leq x \leq \frac{1}{1 - \alpha}, \ 0 \leq u \leq \alpha x \right\}. \tag{4}$$

After the control action is selected, a convex cost function

$$c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$$

is revealed, and the incurred cost is $c_t(x_t, u_t)$. We assume each c_t is convex in (x, u) and satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there exist constants $L_x, L_u > 0$, independent of t, such that for all $(x, u), (x', u') \in \mathcal{D}_\alpha$,

$$|c_t(x, u) - c_t(x', u')| \leq L_x |x - x'| + L_u |u - u'|. \tag{5}$$

We also assume access, at every $(x, u) \in \mathcal{D}_\alpha$, to a cost subgradient

$$\boldsymbol{\zeta}_t(x, u) = \left(\zeta^x_t(x, u), \zeta^u_t(x, u)\right) \in \partial c_t(x, u)$$

satisfying

$$|\zeta^x_t(x, u)| \leq L_x, \qquad |\zeta^u_t(x, u)| \leq L_u.$$

The stated subgradient bounds hold, for example, when (5) extends to an open neighborhood of $\mathcal{D}_\alpha$; see [25, Thm. 9.13].

a) Assumptions:

1) **Retention coefficient:** $\alpha \in (0,1)$ is fixed and known (a fraction $1-\alpha$ of the state is lost each period).[2]
2) **Arrival bounds:** $w_t \in [0,1]$ for all $t \in [T]$. In particular, arrivals are nonnegative resource increments, in contrast to signed process-noise disturbances common in classical control.
3) **Information pattern:** (w_t, c_t) may depend on the history through time $t-1$, but not on the current action or randomization, and is revealed only after u_t is chosen.
4) **Cost regularity:** each c_t is convex in (x,u), obeys (5) on $\mathcal{D}_\alpha$, and admits the bounded subgradient oracle described above.
5) **Feasible policy class:** both the online policy and the benchmark policies are required to satisfy (2) under (3).

Let $\{(x_t,u_t)\}_{t=1}^T$ denote the realized trajectory of the online controller. Define its cumulative cost by

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^T c_t(x_t,u_t). \tag{6}$$

For a fixed positive integer H, let $\mathbf{m}$ range over the SDAC parameter set $\mathcal{M}_H$ (defined in Section IV). For each $\mathbf{m} \in \mathcal{M}_H$, write

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^T c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$$

for the cumulative cost of the induced feasible trajectory under the same arrival sequence. The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_H} J_T(\mathbf{m}). \tag{7}$$

While the online controller may update its SDAC parameters over time, the benchmark in (7) is the best single fixed SDAC policy chosen in hindsight. The objective is to design an online policy that satisfies (2) for all $t \in [T]$ and achieves sublinear regret as defined in (7).

IV. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

Disturbance-Action Control (DAC) combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate losses and efficient online updates. In our scalar storage model, specializing the baseline feedback term to zero gives the representative form

$$u_t = \sum_{i=1}^H \theta_i w_{t-i}, \tag{8}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1,\dots,\theta_H)^\top \in \mathbb{R}^H$ is a linear filter. We allow nonpositive indices in such sums, with the convention $w_\tau \equiv 0$ for $\tau \le 0$.

For our constrained model, an unrestricted filter is insufficient: its action can violate (2) by exceeding αx_t or becoming negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction yields fixed-policy feasibility while preserving convex surrogate losses and tractable online updates.

[2] We focus on $\alpha \in (0,1)$; the case $\alpha \ge 1$ is discussed in Section VII.

B. SDAC Definition

Fix a memory horizon $H \in \mathbb{N}_{>0}$. Let

$$\mathcal{M}_H \triangleq \big\{\mathbf{m} \in \mathbb{R}^H_{\ge 0} : \ \|\mathbf{m}\|_1 \le 1\big\} \tag{9}$$

be the subprobability simplex. For any $\mathbf{m} \in \mathcal{M}_H$, define the policy $\pi_{\mathbf{m}}$ by

$$u_t(\mathbf{m}) \ \triangleq \ \sum_{i=1}^H m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle, \qquad w_\tau \equiv 0 \text{ for } \tau \le 0, \tag{10}$$

where

$$\boldsymbol{\phi}_t^u \triangleq \big(\alpha w_{t-1}, \alpha^2 w_{t-2}, \dots, \alpha^H w_{t-H}\big)^\top \in \mathbb{R}^H, \tag{11}$$

with the same convention $w_\tau \equiv 0$ for $\tau \le 0$.

The induced state trajectory satisfies

$$x_{t+1}(\mathbf{m}) = \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) + w_t, \qquad x_1(\mathbf{m}) = 0. \tag{12}$$

The SDAC policy class is

$$\Pi_H^{\mathrm{SDAC}} \triangleq \{\pi_{\mathbf{m}} : \mathbf{m} \in \mathcal{M}_H\}. \tag{13}$$

Whenever the policy itself is not needed as an abstract object, we minimize directly over $\mathbf{m} \in \mathcal{M}_H$. The weights α^i align the arrival filter with the retention coefficient α in (3).

We now establish the two structural properties that make SDAC useful: the induced control action always satisfies the feasibility constraint (2), and the induced cost is convex and Lipschitz in $\mathbf{m}$, so that regret minimization against the best fixed SDAC policy is an instance of online convex optimization over the simplex. Both properties follow from an explicit decomposition of the state trajectory that isolates its dependence on $\mathbf{m}$.

Recall that we set $w_\tau \equiv 0$ for $\tau \le 0$, which is used in the base-case arguments below.

Lemma 1 (State decomposition under SDAC)**.** *Fix a positive integer H and $\mathbf{m} \in \mathcal{M}_H$. For $i \in \{1,\dots,H\}$, define*

$$r_{t,i} \triangleq \sum_{j=1}^i \alpha^{j-1} w_{t-j}, \qquad r_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j},$$

where the second sum is empty when $t=1$. Then, for every $t \in [T]$,

$$x_t(\mathbf{m}) = \sum_{i=1}^H m_i r_{t,i} + \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t. \tag{14}$$

Define

$$\boldsymbol{\phi}_t^x \triangleq \big(r_{t,1} - r_t, \dots, r_{t,H} - r_t\big)^\top.$$

Then

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m}\rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle.$$

Moreover,

$$0 \le r_{t,i} \le r_t \le \frac{1}{1-\alpha},$$

and $\mathbf{m} \mapsto x_t(\mathbf{m})$ *is affine.*

Proof. The auxiliary quantities satisfy

$$r_{t+1} = \alpha r_t + w_t, \qquad r_1 = 0, \tag{15}$$

and, for every $i \in \{1, \ldots, H\}$,

$$\begin{aligned} \alpha r_{t,i} - \alpha^i w_{t-i} &= \sum_{j=1}^{i-1} \alpha^j w_{t-j} \\ &= r_{t+1,i} - w_t. \end{aligned} \tag{16}$$

For $i = 1$, the sum in the first line is empty.

At $t = 1$, since $w_\tau \equiv 0$ for $\tau \le 0$,

$$\begin{aligned} r_{1,i} &= \sum_{j=1}^{i} \alpha^{j-1} w_{1-j} = 0, \\ r_1 &= 0, \\ x_1(\mathbf{m}) &= 0, \end{aligned}$$

so (14) holds. Suppose it holds at time t. Using (12) and (10),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \alpha \left(\sum_{i=1}^{H} m_i r_{t,i} + \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t \right) \\ &\quad - \sum_{i=1}^{H} m_i \alpha^i w_{t-i} + w_t \\ &= \sum_{i=1}^{H} m_i \big(\alpha r_{t,i} - \alpha^i w_{t-i}\big) \\ &\quad + \alpha\big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t + w_t. \end{aligned} \tag{17}$$

Substituting (15) and (16) into (17),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \sum_{i=1}^{H} m_i \big(r_{t+1,i} - w_t\big) \\ &\quad + \big(1 - \mathbf{1}^\top \mathbf{m}\big)(r_{t+1} - w_t) + w_t \\ &= \sum_{i=1}^{H} m_i r_{t+1,i} + \big(1 - \mathbf{1}^\top \mathbf{m}\big) r_{t+1}, \end{aligned} \tag{18}$$

because the total coefficient of w_t in the first line is $-\mathbf{1}^\top \mathbf{m} - \big(1 - \mathbf{1}^\top \mathbf{m}\big) + 1 = 0$. This proves (14) by induction.

All summands defining $r_{t,i}$ and r_t are nonnegative. Since $w_\tau = 0$ for $\tau \le 0$, $r_{t,i}$ is a truncation of r_t; hence

$$0 \le r_{t,i} \le r_t \le \sum_{j=1}^{\infty} \alpha^{j-1} = \frac{1}{1-\alpha}.$$

Finally,

$$x_t(\mathbf{m}) = r_t + \sum_{i=1}^{H} m_i \big(r_{t,i} - r_t\big) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle,$$

which is affine in $\mathbf{m}$. □

The first consequence of this decomposition is fixed-policy feasibility.

Lemma 2 (Feasibility of SDAC policies)**.** *For every* $\mathbf{m} \in \mathcal{M}_H$*, the trajectory induced by* (10) *and* (12) *satisfies*

$$0 \le u_t(\mathbf{m}) \le \alpha x_t(\mathbf{m}) \qquad \textit{for all } t \in [T]. \tag{19}$$

Proof. Because m_i, α, and w_{t-i} are nonnegative,

$$u_t(\mathbf{m}) = \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \ge 0.$$

For every $i \in \{1, \ldots, H\}$,

$$\alpha r_{t,i} = \sum_{j=1}^{i} \alpha^j w_{t-j} \ge \alpha^i w_{t-i}. \tag{20}$$

By Lemma 1, $1 - \mathbf{1}^\top \mathbf{m} \ge 0$ and $r_t \ge 0$. Therefore,

$$\begin{aligned} \alpha x_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha r_{t,i} + \alpha\big(1 - \mathbf{1}^\top \mathbf{m}\big) r_t \\ &\ge \sum_{i=1}^{H} m_i \alpha r_{t,i} \\ &\ge \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = u_t(\mathbf{m}), \end{aligned} \tag{21}$$

where the second inequality uses (20). □

For fixed arrivals, both coordinates of the SDAC state–action pair are affine in $\mathbf{m}$. The following lemma records the resulting convexity and the constants used by the online update.

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H,$$

and

$$G_\alpha \triangleq \alpha \left(\frac{L_x}{1-\alpha} + L_u \right). \tag{22}$$

Then ℓ_t *is convex on* $\mathcal{M}_H$*, and for all* $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$*,*

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \le G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.$$

Moreover, at every $\mathbf{m} \in \mathcal{M}_H$*, one can select* $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$ *such that* $\|\mathbf{g}_t\|_\infty \le G_\alpha$*.*

Proof. Lemma 1 and (10) give

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle. \tag{23}$$

For every $i \in \{1, \ldots, H\}$, the definitions of $r_{t,i}$ and r_t give

$$\begin{aligned} (\boldsymbol{\phi}_t^x)_i &= r_{t,i} - r_t \\ &= -\sum_{j=i+1}^{t-1} \alpha^{j-1} w_{t-j}, \end{aligned} \tag{24}$$

where the sum is empty when $i \ge t-1$. Hence

$$\begin{aligned} \big|(\boldsymbol{\phi}_t^x)_i\big| &\le \sum_{j=i+1}^{\infty} \alpha^{j-1} = \frac{\alpha^i}{1-\alpha}, \\ 0 \le (\boldsymbol{\phi}_t^u)_i &= \alpha^i w_{t-i} \le \alpha^i. \end{aligned} \tag{25}$$

Taking the largest coordinate, attained in these upper bounds at $i = 1$, yields

$$\|\boldsymbol{\phi}_t^x\|_\infty \le \frac{\alpha}{1-\alpha}, \qquad \|\boldsymbol{\phi}_t^u\|_\infty \le \alpha. \tag{26}$$

Since (23) is affine in $\mathbf{m}$ and c_t is convex, ℓ_t is convex.

By (5) and (23),

$$\begin{aligned} |\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| &\le L_x\big|\langle \boldsymbol{\phi}_t^x, \mathbf{m}-\mathbf{m}'\rangle\big| + L_u\big|\langle \boldsymbol{\phi}_t^u, \mathbf{m}-\mathbf{m}'\rangle\big| \\ &\le \big(L_x\|\boldsymbol{\phi}_t^x\|_\infty + L_u\|\boldsymbol{\phi}_t^u\|_\infty\big)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &\le \alpha\left(\frac{L_x}{1-\alpha} + L_u\right)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &= G_\alpha\|\mathbf{m}-\mathbf{m}'\|_1. \end{aligned} \tag{27}$$

Let (ζ_t^x, ζ_t^u) denote the components of the bounded subgradient $\boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$ from Section III, and set

$$\mathbf{g}_t \triangleq \zeta_t^x\,\boldsymbol{\phi}_t^x + \zeta_t^u\,\boldsymbol{\phi}_t^u. \tag{28}$$

For every $\mathbf{m}' \in \mathcal{M}_H$, the subgradient inequality and (23) yield

$$\begin{aligned} \ell_t(\mathbf{m}') &\ge \ell_t(\mathbf{m}) + \zeta_t^x\langle \boldsymbol{\phi}_t^x, \mathbf{m}'-\mathbf{m}\rangle + \zeta_t^u\langle \boldsymbol{\phi}_t^u, \mathbf{m}'-\mathbf{m}\rangle \\ &= \ell_t(\mathbf{m}) + \langle \mathbf{g}_t, \mathbf{m}'-\mathbf{m}\rangle. \end{aligned}$$

Thus $\mathbf{g}_t \in \partial\ell_t(\mathbf{m})$. For each $i \in \{1,\dots,H\}$, the exact coordinate decay in (25) gives

$$\begin{aligned} |g_{t,i}| &\le |\zeta_t^x|\big|(\boldsymbol{\phi}_t^x)_i\big| + |\zeta_t^u|(\boldsymbol{\phi}_t^u)_i \\ &\le \alpha^i\left(\frac{L_x}{1-\alpha} + L_u\right) \\ &\le \alpha\left(\frac{L_x}{1-\alpha} + L_u\right) = G_\alpha. \end{aligned} \tag{29}$$

Taking the maximum over i proves

$$\|\mathbf{g}_t\|_\infty \le G_\alpha. \tag{30}$$

□

Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

C. Expressiveness: Proportional State-Feedback Approximation

A standard policy class in online control is the family of linear state-feedback policies $u_t = -Kx_t$, and a foundational property of Disturbance-Action Control is that the DAC class *approximates* any strongly stable linear policy, with approximation error decaying geometrically in the memory horizon H [1], [4].

Under our sign convention, positive u_t drains storage, so proportional feedback is written $u_t = kx_t$. In our scalar model, the feasible proportional state-feedback policies are exactly those with gain $k \in [0, \alpha]$: since $x_t \ge 0$, the feasibility constraint $0 \le u_t \le \alpha x_t$ in (2) holds if and only if $k \in [0, \alpha]$. The next lemma gives an explicit approximation within the feasible class Π_H^{SDAC}.

Lemma 4 (SDAC approximation of proportional state feedback)**.** *Fix $k \in [0, \alpha]$ and set $\lambda_k \triangleq \alpha - k$. Let $\big(x_t(k), u_t(k)\big)$ denote the trajectory of the feasible proportional policy $u_t(k) = k\,x_t(k)$, so that*

$$\begin{aligned} x_{t+1}(k) &= \lambda_k\,x_t(k) + w_t, \\ t &\in [T], \qquad x_1(k) = 0. \end{aligned}$$

Define $\mathbf{m}_H^\star(k) \in \mathbb{R}^H$ by

$$\begin{aligned} \big[\mathbf{m}_H^\star(k)\big]_1 &\triangleq \frac{k}{\alpha}, \\ \big[\mathbf{m}_H^\star(k)\big]_i &\triangleq \frac{k\,\lambda_k^{i-1}}{\alpha^i}, \qquad i = 2,\dots,H. \end{aligned}$$

This definition satisfies $\big[\mathbf{m}_H^\star(k)\big]_i\alpha^i = k\lambda_k^{i-1}$ for every $i \in \{1,\dots,H\}$. Then $\mathbf{m}_H^\star(k) \in \mathcal{M}_H$, and if each c_t satisfies (5)*, then for every $\{w_t\} \subset [0,1]$,*

$$\begin{aligned} &\Big|\sum_{t=1}^T c_t\big(x_t(\mathbf{m}_H^\star(k)), u_t(\mathbf{m}_H^\star(k))\big) - \sum_{t=1}^T c_t\big(x_t(k), u_t(k)\big)\Big| \\ &\qquad \le \Big(L_u + \frac{L_x}{1-\alpha}\Big)\frac{k\,\lambda_k^H}{1-\lambda_k}\,T. \end{aligned} \tag{31}$$

The proof is deferred to Appendix A.

Theorem 1 depends on the memory horizon only through $\sqrt{\log(H+1)}$. Combining that bound with Lemma 4 therefore yields a no-regret guarantee against the best feasible proportional state-feedback policy; Corollary 2 makes this statement precise.

D. Expressiveness: Decreasing Allocation Sequences

Energy-harvesting policies for Bernoulli recharges can be described by the sequence of expenditures following each recharge [11], [12]. This suggests an age-based benchmark for the present additive-arrival model: each unit arriving at time s is assigned the same release sequence over times $s+1, s+2, \dots$. Unlike proportional feedback, this sequence need not be geometric.

Define the set of decreasing feasible allocation sequences

$$\mathcal{Q}_\alpha^\downarrow \triangleq \left\{\mathbf{q} = (q_i)_{i\ge 1} : q_1 \ge q_2 \ge \cdots \ge 0, \quad \sum_{i=1}^\infty \frac{q_i}{\alpha^i} \le 1\right\}. \tag{32}$$

For $\mathbf{q} \in \mathcal{Q}_\alpha^\downarrow$, define

$$u_t(\mathbf{q}) \triangleq \sum_{i=1}^{t-1} q_i w_{t-i}, \qquad x_{t+1}(\mathbf{q}) = \alpha x_t(\mathbf{q}) - u_t(\mathbf{q}) + w_t, \quad x_1(\mathbf{q}) = 0. \tag{33}$$

Here $q_i w_s$ is the release at age i assigned to the arrival cohort w_s. The weighted budget in (32) accounts for leakage: reserving q_i units for release at age i consumes q_i/α^i of the cohort's normalized release budget. Let

$$\Pi_\alpha^{\mathrm{alloc}} \triangleq \{\pi_{\mathbf{q}} : \mathbf{q} \in \mathcal{Q}_\alpha^\downarrow\}.$$

Lemma 5 (SDAC approximation of decreasing allocation sequences)**.** *Every policy in $\Pi_\alpha^{\mathrm{alloc}}$ is feasible. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^\downarrow$ and $H \ge 1$, and define*

$$m_{H,i}(\mathbf{q}) \triangleq \frac{q_i}{\alpha^i}, \qquad i \in [H]. \tag{34}$$

Then $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. *With*

$$R_H(\mathbf{q}) \triangleq \sum_{i=H+1}^{\infty} q_i,$$

the corresponding trajectories satisfy, for every arrival sequence in $[0,1]$,

$$0 \le u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) \le R_H(\mathbf{q}), \tag{35}$$

$$0 \le x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}) \le \frac{R_H(\mathbf{q})}{1-\alpha}. \tag{36}$$

Consequently,

$$\left|\sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) - \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\right| \le \left(L_u + \frac{L_x}{1-\alpha}\right) R_H(\mathbf{q}) T \le \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H+1} T. \tag{37}$$

The proof is deferred to Appendix A.

This class contains every feasible proportional policy. Indeed, for $k \in (0,\alpha]$ and $\lambda_k = \alpha - k$, the sequence

$$q_i(k) = k\lambda_k^{i-1}, \qquad i \ge 1, \tag{38}$$

is decreasing and satisfies

$$\sum_{i=1}^{\infty} \frac{q_i(k)}{\alpha^i} = \frac{k}{\alpha}\sum_{j=0}^{\infty}\left(\frac{\lambda_k}{\alpha}\right)^j = 1.$$

Equations (84) and (33) then induce the same action and state trajectories. The zero-gain policy corresponds to $\mathbf{q} = \mathbf{0}$. Thus the proportional policy family is contained in $\Pi_\alpha^{\text{alloc}}$, generally strictly: the allocation class also permits nongeometric decreasing release profiles.

V. Online Algorithm

We now present our online algorithm, which updates SDAC parameters by online mirror descent (OMD) with unnormalized negentropy (equivalently, the exponentiated gradient method) and a fixed step size $\eta > 0$ over the subprobability simplex $\mathcal{M}_H \subset \mathbb{R}^H$ [26]. Let

$$\Psi(\mathbf{m}) \triangleq \sum_{i=1}^{H}\big(m_i \log m_i - m_i\big), \qquad \mathbf{m} \in \mathbb{R}^H_{\ge 0}, \tag{39}$$

denote the unnormalized negentropy (with the convention $0\log 0 \triangleq 0$). Its Bregman divergence is the generalized Kullback–Leibler divergence

$$D_\Psi(\mathbf{m}\|\mathbf{v}) \triangleq \sum_{i=1}^{H}\Big(m_i \log\tfrac{m_i}{v_i} - m_i + v_i\Big), \tag{40}$$

defined for

$$\mathbf{m} \in \mathbb{R}^H_{\ge 0}, \qquad \mathbf{v} \in \mathbb{R}^H_{>0}.$$

For each time $t \in [T]$, let $\mathbf{m}_t \in \mathcal{M}_H$ be the online parameter. We initialize at

$$\mathbf{m}_1 \triangleq \frac{1}{H+1}\mathbf{1} \tag{41}$$

so that $\sup_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \log(H+1)$ (see the proof of Theorem 1). Henceforth, x_t and u_t, without arguments, denote the state and clipped action generated by the online controller.

a) Control-action selection: The controller first forms the nominal SDAC action

$$\widehat{u}_t \triangleq \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \tag{42}$$

and clips it to the admissible interval $[0, \alpha x_t]$:

$$u_t \triangleq \min\{\widehat{u}_t,\, \alpha x_t\}. \tag{43}$$

The state update is

$$x_{t+1} = \alpha x_t - u_t + w_t, \qquad x_1 = 0. \tag{44}$$

By construction, $\widehat{u}_t \ge 0$ and $u_t \in [0, \alpha x_t]$; with $x_1 = 0$ and $w_t \in [0,1]$, induction on (44) yields $x_t \ge 0$ for all $t \in [T+1]$, so the online trajectory satisfies (2).

The scalar correction in (43) also has an exact Bregman-projection interpretation. This interpretation is useful conceptually, but it does not add a projection to the implementation. The positive initialization and multiplicative OMD update keep $\mathbf{m}_t$ strictly positive for every t; see Lemma 8.

Lemma 6 (Availability clipping as a Bregman projection)**.** *At time* t*, define the parameter set compatible with the available state by*

$$\mathcal{F}_t(x_t) \triangleq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \le \alpha x_t\}. \tag{45}$$

For the negentropy Ψ *in* (39)*, let*

$$\mathbf{m}_t^{\text{exec}} \in \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{F}_t(x_t)} D_\Psi(\mathbf{m}\|\mathbf{m}_t). \tag{46}$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}}\rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} = u_t. \tag{47}$$

Thus the projection formulation and the scalar clipping rule induce exactly the same executed action.

Proof. The set $\mathcal{F}_t(x_t)$ is nonempty because it contains $\mathbf{0}$, and it is compact and convex. Since $\mathbf{m}_t$ is strictly positive coordinatewise, $D_\Psi(\cdot\|\mathbf{m}_t)$ is strictly convex on $\mathcal{M}_H$ and is minimized uniquely at $\mathbf{m}_t$ over any feasible set containing $\mathbf{m}_t$. Consequently, if

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle \le \alpha x_t,$$

then $\mathbf{m}_t^{\text{exec}} = \mathbf{m}_t$, which proves (47) in the feasible case.

Suppose instead that $\mathbf{m}_t$ is infeasible. We claim that the availability constraint is active at $\mathbf{m}_t^{\text{exec}}$. If it were slack, then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}}\rangle < \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle. \tag{48}$$

For sufficiently small $\lambda \in (0,1)$, the point

$$\mathbf{m}_\lambda \triangleq (1-\lambda)\mathbf{m}_t^{\text{exec}} + \lambda\mathbf{m}_t$$

would still belong to $\mathcal{F}_t(x_t)$. Strict convexity and $D_\Psi(\mathbf{m}_t\|\mathbf{m}_t) = 0$ would then give

$$\begin{aligned} D_\Psi(\mathbf{m}_\lambda\|\mathbf{m}_t) &< (1-\lambda)D_\Psi(\mathbf{m}_t^{\text{exec}}\|\mathbf{m}_t) \\ &< D_\Psi(\mathbf{m}_t^{\text{exec}}\|\mathbf{m}_t), \end{aligned}$$

contradicting (46). Hence

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\text{exec}}\rangle = \alpha x_t$$

in the infeasible case, which completes the proof. □

Lemma 6 does not define a different physical controller. The implemented rule remains the scalar minimum (43); the full H-dimensional vector $\mathbf{m}_t^{\text{exec}}$ is never calculated because only its scalar induced action is executed. Forming the feature vectors and their inner products, followed by the OMD update below, takes $O(H)$ time and $O(H)$ memory per round. In particular, the next unconstrained OMD update uses the internal iterate $\mathbf{m}_t$, not $\mathbf{m}_t^{\text{exec}}$. Therefore the internal iterates remain on the fixed domain $\mathcal{M}_H$, and the analysis does not acquire a time-varying comparator set. The clipped and projection-interpreted descriptions produce the same actions, states, and internal OMD iterates.

b) Surrogate cost and OMD update: After (w_t, c_t) is revealed, define the surrogate cost by

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H, \tag{49}$$

which is convex and Lipschitz by Lemma 3. Select a subgradient $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$ satisfying $\|\mathbf{g}_t\|_\infty \le G_\alpha$ as in (22), whose existence and construction are given in that lemma. This step uses the revealed cost function and the full-information subgradient oracle from Section III; it is not a bandit-feedback update. We define the OMD update

$$\mathbf{m}_{t+1} \triangleq \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} \Big\{ \eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m} \| \mathbf{m}_t) \Big\}. \tag{50}$$

The regret analysis in Section VI works directly with this optimization-based definition.

c) Closed-form solution: Although (50) is a constrained optimization problem, it is easy to solve analytically because Ψ is separable and the objective is linear in $\mathbf{g}_t$. As shown in Lemma 8, the minimizer in (50) is exactly the result of two elementary steps. First a multiplicative step,

$$\widetilde{m}_{t+1,i} \triangleq m_{t,i} \exp\big(-\eta g_{t,i}\big), \qquad i = 1, \ldots, H, \tag{51}$$

where $g_{t,i}$ is the i-th coordinate of $\mathbf{g}_t$. This is followed by the Bregman projection onto $\mathcal{M}_H$,

$$\mathbf{m}_{t+1} \triangleq \operatorname{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}) = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}, \tag{52}$$

where

$$\operatorname{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) \triangleq \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}})$$

(the scaling is nontrivial only when $\|\widetilde{\mathbf{m}}_{t+1}\|_1 > 1$). This is the form we implement in practice.

VI. Regret Analysis

The following theorem states that the online controller achieves sublinear regret against every fixed SDAC policy $\pi_{\mathbf{m}}$, where $\mathbf{m}$ is any parameter in $\mathcal{M}_H$. The regret proof has two parts: a standard online convex optimization bound for the surrogate costs ℓ_t, and a stability bound showing that the action-clipped online trajectory stays close to the SDAC policy trajectory driven by the same parameters. Some technical lemmas required for the proof are deferred to Appendix A.

Theorem 1 (Regret bound)**.** *Suppose the cost functions satisfy the regularity and bounded-subgradient assumptions of Section III. Then the OMD controller with action clipping from Section V, initialized as in* (41) *and using step size $\eta > 0$, satisfies*

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha, \tag{53}$$

where

$$\Gamma_\alpha \triangleq \frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}, \tag{54}$$

and $G_\alpha = \alpha\big(\frac{L_x}{1-\alpha} + L_u\big)$ as in (22)*. In particular, the choice*

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}$$

gives

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T \log(H+1)}.$$

For fixed $L_x, L_u > 0$, the leading factor obeys

$$\sqrt{\Gamma_\alpha} = \Theta(\alpha) \quad \text{as } \alpha \downarrow 0,$$

and

$$\sqrt{\Gamma_\alpha} = \Theta\Big((1-\alpha)^{-3/2}\Big) \quad \text{as } \alpha \uparrow 1.$$

The next result shows that the $\sqrt{T}$ dependence is unavoidable and quantifies the stability-margin dependence that any causal feasible controller must incur. Its proof is deferred to Appendix A.

Theorem 2 (Minimax SDAC regret lower bound)**.** *Fix $H \ge 1$ and $T \ge 2$. For every causal feasible online controller $\mathcal{A}$, possibly randomized, there exists a deterministic oblivious sequence of arrivals and costs satisfying the assumptions of Section III such that*

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] \ge \frac{1}{2\sqrt{2}} \left[\sum_{t=2}^{T} \alpha^2 \times \left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha} \right)^2 \right]^{1/2}. \tag{55}$$

The expectation is only over the internal randomization of $\mathcal{A}$; the adversarial sequence whose existence is asserted is deterministic and fixed before the interaction.

Corollary 1 (Lower bound beyond the mixing scale)**.** *Define*

$$q_\alpha \triangleq \left\lceil \frac{\log 2}{\log(1/\alpha)} \right\rceil. \tag{56}$$

If $T \ge 2q_\alpha + 2$, then every causal feasible controller $\mathcal{A}$ admits a deterministic oblivious sequence for which

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] \ge \frac{G_\alpha}{8}\sqrt{T}, \tag{57}$$

where $G_\alpha = \alpha(L_x/(1-\alpha) + L_u)$.

For fixed $L_x, L_u > 0$, Corollary 1 matches the $\Theta(\alpha)$ power of the upper bound as $\alpha \downarrow 0$. As $\alpha \uparrow 1$, $q_\alpha = \Theta((1-\alpha)^{-1})$ and the corollary gives

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] = \Omega\left(\frac{\sqrt{T}}{1-\alpha} \right) \tag{58}$$

whenever T is at least the mixing scale. The upper bound in Theorem 1 is $O((1-\alpha)^{-3/2}\sqrt{T \log(H+1)})$, so a factor

$(1-\alpha)^{-1/2}$ in the stability margin remains open, apart from the memory-dependent logarithm. The lower-bound scaling in (58) agrees with the discounted-memory obstruction in online convex optimization with unbounded memory [9]. This is a comparison of rates rather than a reduction: Theorem 2 is proved directly under the scalar state–action loss and storage-feasibility structure of our model.

The mixing-scale qualification is essential for interpreting the large-α asymptotic. If T is fixed while $\alpha \uparrow 1$, then

$$\frac{1-\alpha^{t-2}}{1-\alpha} \longrightarrow t-2,$$

and the right-hand side of (55) converges to the finite quantity

$$\frac{1}{2\sqrt{2}}\left[\sum_{t=2}^{T}\big(L_u+L_x(t-2)\big)^2\right]^{1/2}.$$

Thus the exact finite-horizon formula, rather than the steady-state scaling in (58), applies in this regime.

The two asymptotic regimes improve the coarser analysis that ignores the exact coordinate decay in (25) and the positive-part contraction in (107). That analysis gives a leading factor that remains nonzero as $\alpha \downarrow 0$ and grows as $(1-\alpha)^{-2}$ as $\alpha \uparrow 1$. The projection interpretation in Lemma 6 does not cause this improvement; it only reexpresses the same scalar clipping rule. The sharper rate follows from the gradient and stability estimates.

Combining Theorem 1 with Lemma 4 yields a uniform guarantee against the best feasible proportional state-feedback policy.

Corollary 2 (No-regret against proportional state feedback)**.** *Set*

$$H_T \triangleq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil\right\}.$$

Run the OMD controller with action clipping using memory horizon $H = H_T$ and the step size η^* of Theorem 1. For each $k \in [0,\alpha]$, let $\big(x_t(k), u_t(k)\big)$ denote the trajectory of the proportional policy $u_t(k) = kx_t(k)$. Define

$$\mathrm{Reg}_T^{\mathrm{prop}} \triangleq J_T^{\mathrm{on}} - \min_{k\in[0,\alpha]} J_T(k),$$

where $J_T(k) \triangleq \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)$. Then

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{prop}} &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)}\\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{\alpha^{H_T+1}}{1-\alpha}\,T,\end{aligned}$$

where Γ_α is defined in (54)*. Since $\alpha^{H_T} \le 1/T$, the second term is $O(1)$, while $\log(H_T+1) = O(\log\log T)$ as $T \to \infty$. Consequently,*

$$\mathrm{Reg}_T^{\mathrm{prop}} = O\left(\sqrt{T\log\log T}\right).$$

Thus $\mathrm{Reg}_T^{\mathrm{prop}} = o(T)$ against the best proportional state-feedback policy with $k \in [0,\alpha]$.

Proof. For $\mathbf{m} \in \mathcal{M}_{H_T}$ and $k \in [0,\alpha]$, write

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big),$$
$$J_T(k) \triangleq \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big).$$

Fix $k \in [0,\alpha]$, set $\lambda_k = \alpha - k$, and use the comparator $\mathbf{m}_{H_T}^{\star}(k)$ from Lemma 4. Then

$$\begin{aligned}J_T^{\mathrm{on}} - J_T(k) &= \left(J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_{H_T}} J_T(\mathbf{m})\right)\\ &\quad + \left(\min_{\mathbf{m}\in\mathcal{M}_{H_T}} J_T(\mathbf{m}) - J_T(k)\right)\\ &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\ &\quad + J_T\big(\mathbf{m}_{H_T}^{\star}(k)\big) - J_T(k)\\ &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^{H_T}}{1-\lambda_k}\,T.\end{aligned} \tag{59}$$

The first inequality uses Theorem 1; the second uses Lemma 4. Since $k \le \alpha$, $\lambda_k \le \alpha$, and $1-\lambda_k \ge 1-\alpha$,

$$\frac{k\lambda_k^{H_T}}{1-\lambda_k} \le \frac{\alpha^{H_T+1}}{1-\alpha}. \tag{60}$$

The right-hand side is independent of k. Taking the minimum comparator cost over $k \in [0,\alpha]$ in (59) and applying (60) proves the finite-time bound.

Finally,

$$H_T \ge \frac{\log T}{\log(1/\alpha)} \implies \alpha^{H_T} \le \frac{1}{T}.$$

Also $H_T = \Theta(\log T)$, so $\log(H_T+1) = O(\log\log T)$ as $T \to \infty$. These two estimates give the stated asymptotic rate. ☐

The same memory choice yields a stronger guarantee for the allocation-sequence class, which contains all proportional policies.

Corollary 3 (No-regret against decreasing allocation sequences)**.** *Run the OMD controller with the horizon H_T and step size specified in Corollary 2. For $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$, define*

$$J_T(\mathbf{q}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)$$

and

$$\mathrm{Reg}_T^{\mathrm{alloc}} \triangleq J_T^{\mathrm{on}} - \inf_{\mathbf{q}\in\mathcal{Q}_\alpha^{\downarrow}} J_T(\mathbf{q}).$$

Then

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{alloc}} &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.\end{aligned} \tag{61}$$

Consequently,

$$\mathrm{Reg}_T^{\mathrm{alloc}} = O\left(\sqrt{T\log\log T}\right) = o(T).$$

Since every feasible proportional policy belongs to $\Pi_\alpha^{\mathrm{alloc}}$, this is a stronger comparator guarantee than Corollary 2.

Proof. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$ and use the SDAC comparator $\mathbf{m}_{H_T}(\mathbf{q})$ from Lemma 5. Theorem 1 and (37) give

$$\begin{aligned} J_T^{\text{on}} - J_T(\mathbf{q}) &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right) R_{H_T}(\mathbf{q}) T \\ &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right) \alpha^{H_T+1} T. \end{aligned}$$

The bound is uniform in $\mathbf{q}$, so taking the infimum comparator cost proves (61). Finally, $\alpha^{H_T} \le 1/T$ and $\log(H_T+1) = O(\log\log T)$. □

VII. Extensions

We briefly discuss three possible extensions. Establishing their full regret bounds is left for future work.

a) Retention $\alpha \ge 1$.: For $\alpha \ge 1$, choose a fixed baseline drain Kx_t such that $\bar{\alpha} \triangleq \alpha - K \in (0,1)$, and write $u_t = Kx_t + v_t$. The residual dynamics become $x_{t+1} = \bar{\alpha}x_t - v_t + w_t$. An SDAC filter for v_t should therefore use the powers $\bar{\alpha}^i$, not α^i. If $0 \le v_t \le \bar{\alpha}x_t$, then $0 \le u_t \le (K+\bar{\alpha})x_t = \alpha x_t$. This reduction suggests an extension of the present analysis with transformed cost and Lipschitz constants; we do not pursue the details here.

b) Vector action / multi-task allocation.: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\ge 0}$ that allocates resources across n tasks, subject to $\mathbf{1}^\top \mathbf{u}_t \le \alpha x_t$, so that the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \mathbf{1}^\top \mathbf{u}_t + w_t$, and the cost $c_t(x_t, \mathbf{u}_t)$ is convex in $(x, \mathbf{u})$. The fixed-policy SDAC construction lifts by replacing the filter vector $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n\times H}_{\ge 0}$ whose entries $M_{a,i}$ sum to at most one (a matrix simplex), with each row a corresponding to a task and each column i to a memory lag. OMD can then run over this matrix simplex. If the nominal total allocation exceeds αx_t, the action vector is scaled by a nonnegative factor so that $\mathbf{1}^\top \mathbf{u}_t = \alpha x_t$.

c) Multiple storages with diagonal dynamics.: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\ge 0}$ and evolves as $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$ with $\mathbf{A} = \operatorname{diag}(\alpha_1, \ldots, \alpha_d)$, $\alpha_i \in (0,1)$, arrivals $\mathbf{w}_t \in [0,1]^d$, and per-coordinate constraints $0 \le u_{t,i} \le \alpha_i x_{t,i}$. The dynamics and feasible set then decouple by coordinate, and the natural SDAC class is a product of simplices. For a jointly convex cost, OMD operates jointly on this product set; the optimization separates by coordinate only when the cost is coordinate-separable. Cross-coupled storage systems (non-diagonal $\mathbf{A}$) require additional analysis.

These constructions retain the basic simplex/OMD structure, but their complete feasibility and regret analyses require separate treatment.

VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. Every fixed SDAC policy is feasible by construction. For the time-varying online controller, a Bregman projection maintains the SDAC parameter constraint, while scalar action clipping maintains feasibility of the actual trajectory. The clipping is also exactly the action induced by projecting the current parameter onto the state-dependent availability set, although this second projection need not be computed. Exact feature decay and the contraction of the clipped state update yield $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy, with upper-bound dependence $\Theta(\alpha)$ as $\alpha \downarrow 0$ and $\Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$. The minimax lower bound matches the small-α power. Once the horizon is at least the mixing scale $\Theta((1-\alpha)^{-1})$, it gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ regret as $\alpha \uparrow 1$; hence a square-root stability-margin gap remains open. For fixed T and $\alpha \uparrow 1$, the exact finite-horizon lower bound must be used instead, because the steady-state scale has not yet been reached. The SDAC approximation results further give $O\left(\sqrt{T\log\log T}\right)$ regret against the best decreasing allocation-sequence policy when $H = \Theta(\log T)$. This class contains all feasible proportional state-feedback policies and also permits nongeometric release profiles. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time. Section VII discusses possible extensions to $\alpha \ge 1$, vector allocation, and diagonal multi-storage systems.

Appendix A
Deferred Proofs

This appendix proves the minimax lower bound from Section VI and collects the deferred proofs of the proportional-state-feedback and allocation-sequence approximation guarantees from Section IV, together with the OMD and stability lemmas leading up to the upper regret bound.

Recall that we set $w_\tau \equiv 0$ for $\tau \le 0$, which is used in several base-case arguments below.

We first prove the lower bound. Following the standard random-sign mechanism for online linear-optimization lower bounds [26], the construction embeds a two-comparator online linear problem into the two endpoint SDAC policies $\mathbf{0}$ and $\mathbf{e}_1$.

Proof of Theorem 2. Fix a causal feasible controller $\mathcal{A}$. Let $\xi_1, \ldots, \xi_T$ be independent Rademacher random variables, sampled before the interaction and independently of the internal randomization of $\mathcal{A}$. Consider the oblivious random construction

$$w_t = 1, \qquad c_t(x,u) = C_\alpha + \xi_t(L_x x - L_u u), \tag{62}$$

where

$$C_\alpha \triangleq \frac{L_x + \alpha L_u}{1-\alpha}. \tag{63}$$

Each c_t is affine and hence convex. For any two points in $\mathcal{D}_\alpha$,

$$\begin{aligned} |c_t(x,u) - c_t(x',u')| &= |L_x(x-x') - L_u(u-u')| \\ &\le L_x|x-x'| + L_u|u-u'|. \end{aligned} \tag{64}$$

The constant gradient $\xi_t(L_x, -L_u)$ also obeys the bounded-subgradient assumption. Moreover, for $(x,u) \in \mathcal{D}_\alpha$,

$$\begin{aligned} |L_x x - L_u u| &\le L_x x + L_u u \\ &\le \frac{L_x + \alpha L_u}{1-\alpha} = C_\alpha. \end{aligned} \tag{65}$$

Thus the offset even makes the costs nonnegative on the feasible domain, although it will cancel from regret.

Because $H \geq 1$, both

$$\mathbf{m}^{(0)} \triangleq \mathbf{0}, \qquad \mathbf{m}^{(1)} \triangleq \mathbf{e}_1 \tag{66}$$

belong to $\mathcal{M}_H$. Under the arrivals in (62), the first endpoint has

$$u_t(\mathbf{m}^{(0)}) = 0, \qquad x_t(\mathbf{m}^{(0)}) = \rho_t \triangleq \frac{1-\alpha^{t-1}}{1-\alpha}. \tag{67}$$

Indeed, this follows by unrolling $x_{t+1} = \alpha x_t + 1$ from $x_1 = 0$. At the second endpoint,

$$u_1(\mathbf{m}^{(1)}) = 0, \qquad u_t(\mathbf{m}^{(1)}) = \alpha, \quad t \geq 2. \tag{68}$$

The state satisfies $x_2(\mathbf{m}^{(1)}) = 1$. If it equals one at a time $t \geq 2$, then

$$x_{t+1}(\mathbf{m}^{(1)}) = \alpha - \alpha + 1 = 1.$$

Consequently,

$$x_t(\mathbf{m}^{(1)}) = 1, \qquad u_t(\mathbf{m}^{(1)}) = \alpha, \quad t \geq 2. \tag{69}$$

For $t \geq 2$, the state difference between the endpoints is

$$\begin{aligned} d_t &\triangleq x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)}) \\ &= \rho_t - 1 \\ &= \frac{\alpha(1-\alpha^{t-2})}{1-\alpha}. \end{aligned} \tag{70}$$

Remove the common offset by defining the centered online and endpoint costs

$$\begin{aligned} &\widehat{J}_T^{\mathrm{on}} \triangleq \sum_{t=1}^{T} \xi_t (L_x x_t - L_u u_t), \\ &\widehat{J}_T^{(j)} \triangleq \sum_{t=1}^{T} \xi_t \left(L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)}) \right), \quad j \in \{0,1\}. \end{aligned} \tag{71}$$

At time t, the pair (x_t, u_t) is determined by the past observations and the controller randomization used before c_t is revealed. It is therefore independent of the fresh sign ξ_t. Conditional mean zero gives

$$\begin{aligned} \mathbb{E}[\xi_t (L_x x_t - L_u u_t) \,|\, x_t, u_t] &= 0, \\ \mathbb{E}\left[\widehat{J}_T^{\mathrm{on}}\right] &= 0. \end{aligned} \tag{72}$$

Likewise, the two fixed endpoint trajectories do not depend on the signs, so

$$\mathbb{E}\left[\widehat{J}_T^{(0)}\right] = \mathbb{E}\left[\widehat{J}_T^{(1)}\right] = 0. \tag{73}$$

Since the two endpoints agree at $t = 1$, equations (69)–(70) give

$$\begin{aligned} \widehat{J}_T^{(0)} - \widehat{J}_T^{(1)} &= \sum_{t=2}^{T} \xi_t (L_x d_t + \alpha L_u) \\ &= \sum_{t=2}^{T} \xi_t a_t, \end{aligned} \tag{74}$$

where

$$a_t \triangleq \alpha \left(L_u + \frac{L_x (1-\alpha^{t-2})}{1-\alpha} \right). \tag{75}$$

Minimizing over all of $\mathcal{M}_H$ can only decrease the comparator cost. After canceling the common term TC_α, the regret therefore satisfies pointwise

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \geq \widehat{J}_T^{\mathrm{on}} - \min\left\{ \widehat{J}_T^{(0)}, \widehat{J}_T^{(1)} \right\}. \tag{76}$$

Using $\min\{a,b\} = (a+b-|a-b|)/2$ together with (72) and (73), we obtain

$$\begin{aligned} \mathbb{E}\left[\mathrm{Reg}_T^{\mathrm{SDAC}}\right] &\geq \frac{1}{2}\mathbb{E}\left[\left| \widehat{J}_T^{(0)} - \widehat{J}_T^{(1)} \right|\right] \\ &= \frac{1}{2}\mathbb{E}\left[\left| \sum_{t=2}^{T} \xi_t a_t \right|\right]. \end{aligned} \tag{77}$$

Khintchine's inequality for Rademacher sums implies

$$\mathbb{E}\left[\left| \sum_{t=2}^{T} \xi_t a_t \right|\right] \geq \frac{1}{\sqrt{2}} \left(\sum_{t=2}^{T} a_t^2 \right)^{1/2}. \tag{78}$$

Substitution of (75) into (77)–(78) proves (55) under the joint randomization of the signs and the controller.

It remains to make the adversarial sequence deterministic. For a fixed sign vector $\boldsymbol{\xi} \in \{-1,1\}^T$, let

$$R_{\mathcal{A}}(\boldsymbol{\xi}) \triangleq \mathbb{E}_{\mathcal{A}}\left[\mathrm{Reg}_T^{\mathrm{SDAC}} \,\middle|\, \boldsymbol{\xi}\right].$$

The bound just proved states that the average of $R_{\mathcal{A}}(\boldsymbol{\xi})$ over the 2^T sign vectors is at least the right-hand side of (55). Hence at least one deterministic vector $\boldsymbol{\xi}^\star$ attains that value. Fixing $\boldsymbol{\xi}^\star$ in (62) produces a deterministic oblivious admissible sequence, and the only remaining expectation is over the internal randomization of $\mathcal{A}$. □

Proof of Corollary 1. The definition of q_α gives

$$\alpha^{q_\alpha} \leq \alpha^{\log 2/\log(1/\alpha)} = \frac{1}{2}. \tag{79}$$

Therefore, for every $t \geq q_\alpha + 2$,

$$\begin{aligned} a_t &\geq \alpha \left(L_u + \frac{L_x}{2(1-\alpha)} \right) \\ &\geq \frac{\alpha}{2} \left(L_u + \frac{L_x}{1-\alpha} \right) = \frac{G_\alpha}{2}. \end{aligned} \tag{80}$$

There are $T - q_\alpha - 1$ indices in $\{q_\alpha + 2, \ldots, T\}$. If $T \geq 2q_\alpha + 2$, then

$$T - q_\alpha - 1 \geq \frac{T}{2}. \tag{81}$$

Applying Theorem 2 and retaining only these indices,

$$\begin{aligned} \mathbb{E}_{\mathcal{A}}\left[\mathrm{Reg}_T^{\mathrm{SDAC}}\right] &\geq \frac{1}{2\sqrt{2}} \left[\frac{T}{2} \left(\frac{G_\alpha}{2} \right)^2 \right]^{1/2} \\ &= \frac{G_\alpha}{8}\sqrt{T}. \end{aligned} \tag{82}$$

This is (57). □

We next prove the proportional-policy approximation guarantee stated in Section IV-C.

Proof of Lemma 4. Fix $k \in [0, \alpha]$ and set $\lambda_k \triangleq \alpha - k \in [0, \alpha]$.

Proportional-policy trajectory. Substituting $u_t(k) = k x_t(k)$ into the dynamics gives

$$x_{t+1}(k) = \lambda_k x_t(k) + w_t, \qquad x_1(k) = 0. \tag{83}$$

Unrolling (83),

$$\begin{aligned} x_t(k) &= \sum_{j=1}^{t-1} \lambda_k^{j-1} w_{t-j}, \\ u_t(k) &= k \sum_{j=1}^{t-1} \lambda_k^{j-1} w_{t-j}. \end{aligned} \tag{84}$$

Thus $x_t(k) \geq 0$, and $k \in [0, \alpha]$ implies

$$0 \leq u_t(k) = k x_t(k) \leq \alpha x_t(k).$$

Simplex membership. Each coordinate of $\mathbf{m}_H^\star(k)$ is non-negative. If $k = 0$, then $\mathbf{m}_H^\star(0) = \mathbf{0} \in \mathcal{M}_H$. If $k = \alpha$, then $\lambda_k = 0$, so $\big[\mathbf{m}_H^\star(\alpha)\big]_1 = 1$ and $\big[\mathbf{m}_H^\star(\alpha)\big]_i = 0$ for $i = 2, \ldots, H$, hence $\mathbf{m}_H^\star(\alpha) \in \mathcal{M}_H$. Now suppose $0 < k < \alpha$. Since $k = \alpha - \lambda_k$, one has $k/\alpha = 1 - \lambda_k/\alpha$, and $\lambda_k/\alpha \in (0, 1)$. Consequently,

$$\begin{aligned} \sum_{i=1}^{H} \big[\mathbf{m}_H^\star(k)\big]_i &= \frac{k}{\alpha} + \sum_{i=2}^{H} \frac{k\,\lambda_k^{i-1}}{\alpha^i} \\ &= \frac{k}{\alpha} \sum_{j=0}^{H-1} \left(\frac{\lambda_k}{\alpha}\right)^j \\ &= \left(1 - \frac{\lambda_k}{\alpha}\right) \sum_{j=0}^{H-1} \left(\frac{\lambda_k}{\alpha}\right)^j \\ &= 1 - \left(\frac{\lambda_k}{\alpha}\right)^H \leq 1. \end{aligned} \tag{85}$$

Hence $\mathbf{m}_H^\star(k) \in \mathcal{M}_H$.

Action approximation. The identity $\big[\mathbf{m}_H^\star(k)\big]_i \alpha^i = k \lambda_k^{i-1}$ (with $\lambda_k^0 \triangleq 1$) gives

$$u_t\big(\mathbf{m}_H^\star(k)\big) = k \sum_{i=1}^{H} \lambda_k^{i-1} w_{t-i}. \tag{86}$$

Define

$$e_t^u \triangleq u_t(k) - u_t\big(\mathbf{m}_H^\star(k)\big).$$

Using $w_\tau = 0$ for $\tau \leq 0$ in (84) and (86),

$$e_t^u = k \sum_{j=H+1}^{t-1} \lambda_k^{j-1} w_{t-j}, \tag{87}$$

where the sum is empty when $t \leq H + 1$. Therefore,

$$\begin{aligned} 0 \leq e_t^u &\leq k \sum_{j=H+1}^{\infty} \lambda_k^{j-1} \\ &= \frac{k \lambda_k^H}{1 - \lambda_k}. \end{aligned} \tag{88}$$

State approximation. Let

$$e_t^x \triangleq x_t\big(\mathbf{m}_H^\star(k)\big) - x_t(k).$$

Subtracting the two state recursions yields

$$e_{t+1}^x = \alpha e_t^x + e_t^u, \qquad e_1^x = 0. \tag{89}$$

Hence

$$e_t^x = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s^u. \tag{90}$$

Combining (88) and (90),

$$\begin{aligned} 0 \leq e_t^x &\leq \frac{k \lambda_k^H}{1 - \lambda_k} \sum_{s=1}^{t-1} \alpha^{t-1-s} \\ &\leq \frac{k \lambda_k^H}{(1 - \lambda_k)(1 - \alpha)}. \end{aligned} \tag{91}$$

Cost approximation. By (5), (88), and (91),

$$\begin{aligned} &\big| c_t\big(x_t(\mathbf{m}_H^\star(k)), u_t(\mathbf{m}_H^\star(k))\big) - c_t\big(x_t(k), u_t(k)\big) \big| \\ &\quad \leq L_x e_t^x + L_u e_t^u \\ &\quad \leq \frac{L_x k \lambda_k^H}{(1 - \lambda_k)(1 - \alpha)} + \frac{L_u k \lambda_k^H}{1 - \lambda_k} \\ &\quad = \left(L_u + \frac{L_x}{1 - \alpha}\right) \frac{k \lambda_k^H}{1 - \lambda_k}. \end{aligned} \tag{92}$$

Summing (92) over $t = 1, \ldots, T$ and applying the triangle inequality proves (31). $\square$

Proof of Lemma 5. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^\downarrow$ and set

$$a_i \triangleq \frac{q_i}{\alpha^i}, \qquad b_j \triangleq 1 - \sum_{i=1}^{j-1} a_i, \qquad j \geq 1.$$

The weighted-budget constraint gives $a_i \geq 0$ and $b_j \geq 0$. Moreover,

$$x_t(\mathbf{q}) = \sum_{j=1}^{t-1} \alpha^{j-1} b_j w_{t-j}. \tag{93}$$

Indeed, the identity is immediate at $t = 1$. If it holds at time t, then

$$\begin{aligned} x_{t+1}(\mathbf{q}) &= w_t + \sum_{j=1}^{t-1} \big(\alpha^j b_j - q_j\big) w_{t-j} \\ &= w_t + \sum_{j=1}^{t-1} \alpha^j b_{j+1} w_{t-j}, \end{aligned}$$

because $q_j = \alpha^j a_j$ and $b_{j+1} = b_j - a_j$. Reindexing gives (93) at time $t + 1$.

For each $j \geq 1$,

$$a_j \leq 1 - \sum_{i=1}^{j-1} a_i = b_j.$$

Consequently,

$$\begin{aligned} 0 \leq u_t(\mathbf{q}) &= \sum_{j=1}^{t-1} \alpha^j a_j w_{t-j} \\ &\leq \sum_{j=1}^{t-1} \alpha^j b_j w_{t-j} = \alpha x_t(\mathbf{q}), \end{aligned}$$

which proves feasibility.

For the truncated vector in (34),

$$m_{H,i}(\mathbf{q}) = a_i \geq 0, \qquad \sum_{i=1}^{H} m_{H,i}(\mathbf{q}) \leq \sum_{i=1}^{\infty} a_i \leq 1.$$

Thus $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. Its SDAC action is

$$u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=1}^{\min\{H, t-1\}} q_i w_{t-i}.$$

It follows that

$$0 \le u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=H+1}^{t-1} q_i w_{t-i} \le R_H(\mathbf{q}),$$

where the middle sum is empty when $t \le H+1$. This proves (35).

Define

$$e_t^x \triangleq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}), \qquad e_t^u \triangleq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big).$$

The two state recursions give

$$e_{t+1}^x = \alpha e_t^x + e_t^u, \qquad e_1^x = 0.$$

Since $0 \le e_t^u \le R_H(\mathbf{q})$,

$$0 \le e_t^x = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s^u \le \frac{R_H(\mathbf{q})}{1-\alpha},$$

which proves (36). The Lipschitz property now yields

$$\big|c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) - c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\big| \le \left(L_u + \frac{L_x}{1-\alpha}\right) R_H(\mathbf{q}).$$

Finally,

$$\begin{aligned} R_H(\mathbf{q}) &= \sum_{i=H+1}^{\infty} \alpha^i a_i \\ &\le \alpha^{H+1} \sum_{i=H+1}^{\infty} a_i \le \alpha^{H+1}. \end{aligned}$$

Summing the per-period cost bound proves (37). $\square$

The remaining lemmas establish the OMD and stability estimates used in the proof of Theorem 1; they rely on the state-decomposition, feasibility, and surrogate-cost lemmas already established in Section IV (Lemmas 1, 2, and 3).

Lemma 7 (Bregman projection onto $\mathcal{M}_H$)**.** *Let D_Ψ be the Bregman divergence* (40) *of the unnormalized negentropy* (39)*. Fix $\widetilde{\mathbf{m}} \in \mathbb{R}^H_{>0}$. Then the Bregman projection of $\widetilde{\mathbf{m}}$ onto $\mathcal{M}_H$,*

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) \triangleq \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}),$$

is given by

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\max\{1, \|\widetilde{\mathbf{m}}\|_1\}}.$$

Proof. The projection objective $\mathbf{m} \mapsto D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}})$ is strictly convex, and, for $m_i > 0$,

$$\frac{\partial}{\partial m_i} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}) = \log \frac{m_i}{\widetilde{m}_i}. \tag{94}$$

Its unconstrained minimizer is therefore $\mathbf{m} = \widetilde{\mathbf{m}}$.

If $\|\widetilde{\mathbf{m}}\|_1 \le 1$, this minimizer belongs to $\mathcal{M}_H$, so $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \widetilde{\mathbf{m}}$. Suppose $\|\widetilde{\mathbf{m}}\|_1 > 1$. The constraint $\|\mathbf{m}\|_1 \le 1$ is then active. Moreover, the minimizer has no zero coordinate: if $m_\ell = 0$, transferring an arbitrarily small amount of mass from a positive coordinate to coordinate ℓ has directional derivative $-\infty$, because

$$\lim_{z\downarrow 0} \log \frac{z}{\widetilde{m}_\ell} = -\infty.$$

Thus the nonnegativity constraints are inactive. With $\nu \ge 0$ denoting the multiplier of $\|\mathbf{m}\|_1 \le 1$, stationarity and complementary slackness give

$$\log \frac{m_i}{\widetilde{m}_i} + \nu = 0, \qquad \|\mathbf{m}\|_1 = 1. \tag{95}$$

The first relation in (95) implies

$$m_i = \widetilde{m}_i e^{-\nu}.$$

Summing over i and using the second relation in (95),

$$1 = \|\widetilde{\mathbf{m}}\|_1 e^{-\nu}, \qquad e^{-\nu} = \frac{1}{\|\widetilde{\mathbf{m}}\|_1}.$$

Hence $\nu = \log \|\widetilde{\mathbf{m}}\|_1 > 0$ and

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\|\widetilde{\mathbf{m}}\|_1}.$$

Combining the cases $\|\widetilde{\mathbf{m}}\|_1 \le 1$ and $\|\widetilde{\mathbf{m}}\|_1 > 1$ proves the stated formula. $\square$

Lemma 8 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$, and let $\widetilde{\mathbf{m}}_{t+1}$ be defined coordinatewise by* (51)*, i.e. $\widetilde{m}_{t+1,i} = m_{t,i} e^{-\eta g_{t,i}}$. Then*

$$\begin{aligned} &\operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} \Big\{ \eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t) \Big\} \\ &\quad = \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) = \mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}), \end{aligned}$$

where $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}$ is the Bregman projection of Lemma 7. Consequently, the OMD update (50) *coincides exactly with the closed-form OMD update* (51)–(52)*, and in particular $\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$. Thus the positive initialization* (41) *keeps every algorithm iterate strictly positive.*

Proof. Fix $\mathbf{m} \in \mathbb{R}^H_{\ge 0}$ and a coordinate i. Then

$$\begin{aligned} \eta g_{t,i}\, m_i + m_i \log \frac{m_i}{m_{t,i}} &= m_i \left(\eta g_{t,i} + \log \frac{m_i}{m_{t,i}} \right) \\ &= m_i \log \frac{m_i}{\widetilde{m}_{t+1,i}}. \end{aligned} \tag{96}$$

Substitution into the OMD objective gives

$$\begin{aligned} &\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t) \\ &\quad = D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^{H} (m_{t,i} - \widetilde{m}_{t+1,i}). \end{aligned} \tag{97}$$

The final sum is independent of $\mathbf{m}$. Hence the two optimization problems have the same minimizer, and Lemma 7 gives (52). Since $\mathbf{m}_t > 0$ coordinatewise, both the multiplicative step and the subsequent positive rescaling preserve strict positivity. $\square$

Lemma 9 (One-step movement of OMD)**.** *For the OMD update with step size $\eta > 0$ and subgradient $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$,*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \le \eta\, \|\mathbf{g}_t\|_\infty.$$

In particular, since $\|\mathbf{g}_t\|_\infty \le G_\alpha$, $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \le \eta G_\alpha$.

Proof. First-order optimality of (50), evaluated at the feasible point $\mathbf{m}_t$, gives

$$\begin{aligned}&\langle \eta\mathbf{g}_t + \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \geq 0,\\ &D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) + D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1})\\ &\quad\leq \eta\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &\quad\leq \eta\|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \tag{98}$$

The second line uses the symmetric Bregman identity.

For every $\mathbf{z} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$ and $\mathbf{v} \in \mathbb{R}^H$,

$$\begin{aligned}\mathbf{v}^\top \nabla^2\Psi(\mathbf{z})\mathbf{v} &= \sum_{i=1}^H \frac{v_i^2}{z_i}\\ &\geq \frac{\|\mathbf{v}\|_1^2}{\sum_{i=1}^H z_i}\\ &\geq \|\mathbf{v}\|_1^2.\end{aligned} \tag{99}$$

The first inequality is Cauchy–Schwarz. The segment joining $\mathbf{m}_t$ and $\mathbf{m}_{t+1}$ lies in $\mathcal{M}_H \cap \mathbb{R}^H_{>0}$ by Lemma 8; integrating (99) along this segment yields

$$D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) \geq \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2,$$
$$D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1}) \geq \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2.$$

Combining these inequalities with (98),

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2 \leq \eta\|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.$$

The claim is immediate if $\mathbf{m}_{t+1} = \mathbf{m}_t$; otherwise divide by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$. □

Lemma 10 (OMD regret for the surrogate losses)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (50) *satisfies*

$$\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \tag{100}$$

Proof. Convexity of ℓ_t gives

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}\rangle\\ &= \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle.\end{aligned} \tag{101}$$

First-order optimality of (50), evaluated at $\mathbf{m} \in \mathcal{M}_H$, yields

$$\begin{aligned}\eta\langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle &\leq \langle \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m} - \mathbf{m}_{t+1}\rangle\\ &= D_\Psi(\mathbf{m}\|\mathbf{m}_t) - D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})\\ &\quad - D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t).\end{aligned} \tag{102}$$

The equality is the three-point Bregman identity. Substituting (102) into (101),

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) \leq{}& \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})}{\eta}\\ &+ \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &- \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}.\end{aligned} \tag{103}$$

The strong-convexity calculation (99) and Hölder's inequality give

$$\begin{aligned}&\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}\\ &\quad\leq \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta}\\ &\quad\leq \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2.\end{aligned} \tag{104}$$

The last inequality is Young's inequality. Combining (103) and (104), and then summing over t, gives

$$\begin{aligned}\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \leq{}& \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta}\\ &- \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})}{\eta}\\ &+ \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2.\end{aligned}$$

Since $D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1}) \geq 0$, this proves (100). □

Lemma 11 (Positive-part representation of the dynamics)**.** *For $\mathbf{m} \in \mathcal{M}_H$, define*

$$F_t(x; \mathbf{m}) \triangleq [\alpha x - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle]_+ + w_t. \tag{105}$$

Then the clipped online state and every fixed-policy state satisfy

$$x_{t+1} = F_t(x_t; \mathbf{m}_t), \qquad x_{t+1}(\mathbf{m}) = F_t(x_t(\mathbf{m}); \mathbf{m}). \tag{106}$$

Moreover, for all $x, x' \geq 0$ and $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$\begin{aligned}&|F_t(x; \mathbf{m}) - F_t(x'; \mathbf{m}')|\\ &\quad\leq |\alpha(x - x') - \langle \boldsymbol{\phi}_t^u, \mathbf{m} - \mathbf{m}'\rangle|\\ &\quad\leq \alpha|x - x'| + \|\boldsymbol{\phi}_t^u\|_\infty \|\mathbf{m} - \mathbf{m}'\|_1.\end{aligned} \tag{107}$$

Proof. For the online trajectory, (43) implies

$$\begin{aligned}\alpha x_t - u_t &= \alpha x_t - \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\}\\ &= [\alpha x_t - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle]_+.\end{aligned} \tag{108}$$

For a fixed $\mathbf{m}$, Lemma 2 gives

$$\alpha x_t(\mathbf{m}) - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \geq 0,$$

so inserting a positive part does not change its state update. This proves (106). Finally, the map $z \mapsto [z]_+$ is nonexpansive:

$$|[z]_+ - [z']_+| \leq |z - z'|.$$

Apply this inequality to the two arguments in (105), and then apply Hölder's inequality, to obtain (107). □

Lemma 12 (Contractive state and action deviation)**.** *For the clipped online trajectory and the counterfactual fixed-policy trajectory associated with the current internal iterate, one has, for every $t \in [T]$,*

$$|x_t - x_t(\mathbf{m}_t)| \leq \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \tag{109}$$

$$0 \leq u_t(\mathbf{m}_t) - u_t \leq \frac{\alpha^2\eta G_\alpha}{(1-\alpha)^2}. \tag{110}$$

Here $x_t(\mathbf{m}_t)$ is generated by using the single fixed parameter $\mathbf{m}_t$ from time 1 through time t.

Proof. Define
$$d_t \triangleq |x_t - x_t(\mathbf{m}_t)|.$$

Since $x_1 = x_1(\mathbf{m}_1) = 0$, one has $d_1 = 0$. For $t \in \{1,\ldots,T-1\}$, Lemma 11 gives

$$\begin{aligned} d_{t+1} &= |F_t(x_t;\mathbf{m}_t) - F_t(x_t(\mathbf{m}_{t+1});\mathbf{m}_{t+1})| \\ &\le \alpha\,|x_t - x_t(\mathbf{m}_{t+1})| + \|\boldsymbol{\phi}_t^u\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{111}$$

The affine state representation (23) implies

$$\begin{aligned} |x_t(\mathbf{m}_t) - x_t(\mathbf{m}_{t+1})| &= |\langle \boldsymbol{\phi}_t^x, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle| \\ &\le \|\boldsymbol{\phi}_t^x\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{112}$$

Combining the triangle inequality with (111) and (112) yields

$$\begin{aligned} d_{t+1} &\le \alpha d_t \\ &\quad + (\alpha\|\boldsymbol{\phi}_t^x\|_\infty + \|\boldsymbol{\phi}_t^u\|_\infty)\,\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{113}$$

By (26),

$$\begin{aligned} \alpha\|\boldsymbol{\phi}_t^x\|_\infty + \|\boldsymbol{\phi}_t^u\|_\infty &\le \frac{\alpha^2}{1-\alpha} + \alpha \\ &= \frac{\alpha}{1-\alpha}. \end{aligned} \tag{114}$$

Lemma 9 and $\|\mathbf{q}_t\|_\infty \le G_\alpha$ therefore reduce (113) to

$$d_{t+1} \le \alpha d_t + \frac{\alpha\eta G_\alpha}{1-\alpha}. \tag{115}$$

Unrolling from $d_1 = 0$ gives

$$\begin{aligned} d_t &\le \frac{\alpha\eta G_\alpha}{1-\alpha}\sum_{j=0}^{t-2}\alpha^j \\ &\le \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \end{aligned} \tag{116}$$

with an empty sum when $t = 1$. This proves (109).

Finally, $\widehat{u}_t = u_t(\mathbf{m}_t)$ and fixed-policy feasibility imply

$$\begin{aligned} 0 \le u_t(\mathbf{m}_t) - u_t &= [u_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\le [\alpha x_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\le \alpha d_t. \end{aligned} \tag{117}$$

Combining (117) with (109) proves (110). □

Lemma 13 (Stability term)**.** *For all horizons T,*

$$\sum_{t=1}^{T}\Big(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\Big) \le \frac{\alpha\eta T G_\alpha(L_x + \alpha L_u)}{(1-\alpha)^2}.$$

Proof. Both state–action pairs lie in $\mathcal{D}_\alpha$. Hence (5) and Lemma 12 give

$$\begin{aligned} &c_t(x_t,u_t) - \ell_t(\mathbf{m}_t) \\ &\quad \le L_x\big|x_t - x_t(\mathbf{m}_t)\big| + L_u\big|u_t - u_t(\mathbf{m}_t)\big| \\ &\quad \le \frac{\alpha\eta G_\alpha L_x}{(1-\alpha)^2} + \frac{\alpha^2\eta G_\alpha L_u}{(1-\alpha)^2} \\ &\quad = \frac{\alpha\eta G_\alpha(L_x + \alpha L_u)}{(1-\alpha)^2}. \end{aligned} \tag{118}$$

Summing (118) over $t = 1,\ldots,T$ proves the claim. □

Proof of Theorem 1. Define the stability term

$$\mathrm{Stab}_T \triangleq \sum_{t=1}^{T}\left(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\right).$$

Using $\ell_t(\mathbf{m}) = c_t(x_t(\mathbf{m}), u_t(\mathbf{m}))$, the regret decomposes as

$$\mathrm{Reg}_T^{\mathrm{SDAC}} = \mathrm{Stab}_T + \sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}). \tag{119}$$

We first bound the initial Bregman divergence. Fix $\mathbf{m} \in \mathcal{M}_H$. Since $\mathbf{m}_1 = \frac{1}{H+1}\mathbf{1}$,

$$\begin{aligned} D_\Psi(\mathbf{m}\|\mathbf{m}_1) &= \sum_{i=1}^{H} m_i \log\big((H+1)m_i\big) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1} \\ &= \sum_{i=1}^{H} m_i \log m_i + \|\mathbf{m}\|_1 \log(H+1) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \end{aligned} \tag{120}$$

Because $z\log z \le 0$ for $z \in [0,1]$,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \|\mathbf{m}\|_1 \log(H+1) - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \tag{121}$$

The right-hand side is affine in $\|\mathbf{m}\|_1 \in [0,1]$, and its values at the endpoints are

$$\begin{aligned} \frac{H}{H+1} &\le \log(H+1), \\ \log(H+1) - \frac{1}{H+1} &\le \log(H+1). \end{aligned}$$

Here the first inequality follows from $\log z \ge 1 - z^{-1}$ for $z \ge 1$. Therefore,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \log(H+1) \qquad \text{for every } \mathbf{m} \in \mathcal{M}_H. \tag{122}$$

Lemma 10, $\|\mathbf{g}_t\|_\infty \le G_\alpha$, and (122) imply, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned} \sum_{t=1}^{T}\big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\ &\le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}. \end{aligned} \tag{123}$$

Since (123) holds uniformly over $\mathbf{m}$,

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}) \le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}. \tag{124}$$

Lemma 13 gives

$$\mathrm{Stab}_T \le \frac{\alpha\eta T G_\alpha(L_x + \alpha L_u)}{(1-\alpha)^2}. \tag{125}$$

Substituting (124) and (125) into (119),

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{SDAC}} &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2} \\ &\quad + \frac{\alpha\eta T G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2} \\ &= \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha.\end{aligned} \tag{126}$$

The right-hand side of (126) is positive for every $\eta > 0$. Differentiating with respect to η yields

$$-\frac{\log(H+1)}{\eta^2} + T\Gamma_\alpha.$$

The unique minimizer is therefore

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}. \tag{127}$$

At this value the two summands in (126) are equal, so

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T \log(H+1)}. \tag{128}$$

It remains to verify the dependence on α. As $\alpha \downarrow 0$,

$$\begin{aligned}\frac{G_\alpha}{\alpha} &= \frac{L_x}{1-\alpha} + L_u \longrightarrow L_x + L_u, \\ \frac{\Gamma_\alpha}{\alpha^2} &= \frac{1}{2}\left(\frac{G_\alpha}{\alpha}\right)^2 + \frac{G_\alpha}{\alpha}\frac{L_x + \alpha L_u}{(1-\alpha)^2} \\ &\longrightarrow \frac{(L_x + L_u)^2}{2} + L_x(L_x + L_u) > 0.\end{aligned}$$

Therefore $\sqrt{\Gamma_\alpha} = \Theta(\alpha)$ as $\alpha \downarrow 0$. As $\alpha \uparrow 1$, write $\varepsilon = 1 - \alpha$. Then

$$\begin{aligned}\varepsilon G_\alpha &= \alpha(L_x + \varepsilon L_u) \longrightarrow L_x, \\ \varepsilon^3 \Gamma_\alpha &= \frac{\varepsilon^3 G_\alpha^2}{2} + \alpha(\varepsilon G_\alpha)(L_x + \alpha L_u) \\ &\longrightarrow L_x(L_x + L_u) > 0.\end{aligned}$$

Thus $\sqrt{\Gamma_\alpha} = \Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$. □

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